

6(a)	Apply Charles' law $\frac{V_1}{T_1} = \frac{V_2}{T_2}$ $V_2 = \frac{T_2 V_1}{T_1} = \frac{(273.15 + 58)(52.1)}{(273.15 + 41)}$
	$V_2 = 54.9 \text{ cm}^3$
(b)	The real gas does not behave like an ideal gas.
(c)	Pressure: Piston is free to move in and out
	Amount of gas: cap stops gas leaving or entering the syringe and the piston seals the gas
(d)	Increase in T causes the molecules to move with higher kinetic energy
	The particle will exert a larger force on the wall as the (rate of) change in momentum when the particle collide with the wall will be larger.
	To ensure pressure stays the same, V must increase so that the frequency of the collisions must decrease

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7)a)	Rate of decay
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